

MARCELO VINICIUS TAPARELLI

AI Engineer | Software Engineer

Python · RAG · Agents · Evals · TypeScript · Bun · Node.js · PostgreSQL · AWS

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PROFESSIONAL SUMMARY

AI Engineer with a strong Software Engineering foundation and experience in backend, APIs, and production software. I build AI systems with RAG, agents, evals, model evaluation, and human review. From business problem to production, a Product perspective guides architecture, verification, security, observability, and measurement.

PROFESSIONAL EXPERIENCE

Agência Catus | Full Stack Developer | Jul 2026 - Present

- Develop and evolve the agency's new website and two WordPress blogs, including Custom Post Types, theme customization, and new features.
- Maintain an e-commerce implementation on the Wake platform, troubleshoot production issues, and work on performance/Core Web Vitals.
- Deploy directly to servers and continuously improve web applications.

EVAG Comunicação e Tecnologia | Developer | Apr 2024 - Present

- Developed Google Drive-to-WordPress automation for generating petition pages, reducing manual steps and making the publishing workflow more direct.
- Developed an internal request management solution integrated with GitLab, supporting organization, traceability, and flow of operational demands.
- Develop and maintain systems, websites, and automation with Laravel, PHP, Python, JavaScript, WordPress, and Tailwind, including production fixes and new features.

EVAG Comunicação e Tecnologia | Product Owner | Apr 2023 - Mar 2024

- Managed website delivery from discovery through delivery, connecting clients, design, and development.
- Structured requirements and support workflows, prioritizing across business needs, user experience, and the team's technical capacity.

TECHNICAL SKILLS

AI Engineering: Python, FastAPI, RAG, embeddings, pgvector, hybrid retrieval, LangGraph, OpenAI, agents, evals, structured outputs, guardrails, human-in-the-loop; experimental model evaluation/adaptation with PyTorch and CUDA.

Software Engineering: TypeScript, Bun, Node.js, REST APIs, PostgreSQL, Redis, Git, Clean Architecture, DDD principles, TDD, unit/integration/E2E tests, idempotency and resilience.

Production, security & reliability: OpenTelemetry, authentication, authorization, performance, Docker, Terraform, CI/CD and a temporary non-production AWS lab.

Product perspective: Business problems, requirements, constraints, trade-offs, outcome measurement and deciding when to use AI.

SELECTED PUBLIC PROJECTS

OpsPilot AI: Python/FastAPI, RAG, pgvector, LangGraph/OpenAI. Deterministic policies, human approval, GitLab reconciliation, evals and OpenTelemetry. Independent OpenAI and GitLab sandbox smokes; validated Terraform, no AWS deployment.

Ops Triage AI: Deterministic baseline, local LLM, hybrid policy and human review. Jev/Laya evaluation-only on a frozen synthetic held-out (70 tickets). Laya with PyTorch/CUDA: exact tuple 11.43% zero-shot → 85.71% adapted; validation-based selection. HIGH/CRITICAL recall regressed from 92.86% to 85.71%.

Resilient Transaction API: TypeScript/Bun, PostgreSQL, Redis, idempotency, retries, circuit breaker, observability and Docker. Terraform and a temporary non-production AWS lab with functional validation, destroy and audit.

Salus: Node.js/Express, PostgreSQL/Prisma, Clean Architecture, domain modeling and a manual TDD core; JWT, ownership-based authorization and tests.

EDUCATION

Postgraduate Program in AI Engineering — Cruzeiro do Sul

In progress · expected completion: Apr 2027

Systems Analysis and Development — UniBF

Completed in 2026

Software Engineering — Alura

Back-end · Node.js APIs · Authentication · Testing · Security · DevOps/CI/CD · Cloud/AWS · Secure Development

LANGUAGES

Portuguese — native · English — advanced